

Melinda A. Tran

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EDUCATION

Boston University - College of Engineering

B.S. Computer Engineering, Computer Science Minor, Machine Learning Concentration

Expected May 2027

Relevant Coursework: Machine Learning, Algorithms & Data Structures, Signals & Systems, Computer Org, Computer Networking, Digital Logic, AI

Member: Theta Tau, SWE, uWISE, High Performance Computing Club, EngBrigades

WORK EXPERIENCE

Applied Machine Learning Extern

Chewy

Aug 2026 - Present

Boston, MA

- Built an NLP classification pipeline using TF-IDF, logistic regression, and Keras neural networks to detect AI-generated text across five domains
- Evaluated models using macro-F1, accuracy, precision, recall, confusion matrices, and per-domain error analysis to identify failure modes
- Tested robustness against paraphrased AI-generated text and deploying a Streamlit detector with confidence-based predictions

Software Engineer Intern

Red Hat

May 2026 - Aug 2026

Boston, MA

- Built distributed PyTorch debugging infrastructure in Python and C++, extending c10d, NCCL, and Flight Recorder tooling to diagnose failures, synchronization issues, and communication bottlenecks across multi-GPU training workloads
- Developed per-rank GPU memory visualization by converting CUDA allocation/free events into Chrome Trace Event JSON and integrating Peretto timelines to analyze memory usage, fragmentation, allocation lifetimes, peak usage, and OOM failures across distributed ranks
- Ported torchcomms' Flight Recorder into PyTorch c10d with CUDA event tracking, async retirement, collective metadata capture, and rank-specific traceback collection across 3 matching paths; validated functionality across DDP, FSDP, and distributed collective operations on **2+ GPUs**
- Implemented a backend-agnostic NaN/Inf detection hook for distributed collectives, passing 7 automated tests across Gloo and NCCL backends

Computer Vision Undergraduate Researcher

BU Robotics Lab

Oct 2025 - Sep 2026

Boston, MA

- Integrated neuromorphic maps (sight) with autonomous control systems to automate complex robot manipulation tasks
- Engineered real-time perception-to-control pipelines for autonomous robotic manipulation using MoCap-based pose tracking, ROS message passing, coordinate frame transformations, trajectory generation, and closed-loop feedback control
- Optimized robot control frequency and system performance to improve the accuracy, responsiveness, and efficiency of autonomous tasks by **33%**

LLM Undergraduate Researcher

BIT Lab

Jun 2025 - Nov 2025

Boston, MA

- Investigated LLM architectures, training objectives, fine-tuning, prompting, and memory mechanisms for reasoning and long-context tasks
- Evaluated optimization and training strategies to measure effects on model performance, memorization, and generalization
- Analyzed LLM inference behavior and failure modes to identify patterns in reasoning, context retention, and response quality

Machine Vision Research Intern

AgAID + Cornell University

May 2025 - Aug 2025

Ithaca, NY

- Built and validated CVAT datasets (**100+ images**) to train computer vision models for obstacle detection and autonomous navigation
- Integrated AI perception models with robotic platforms using ROS 2, OpenCV, and Roboflow, achieving **86% accuracy** in orchard field testing
- Developed AI-based traversability pipelines to evaluate terrain navigability and improve autonomous decision-making in agricultural environments

PROJECTS

Machine Learning & Artificial Intelligence Fellow

Break Through Tech + Cornell University | Python | NumPy |

Mar 2026 - Present

Ithaca, NY

- Built end-to-end ML pipelines in Python using pandas, NumPy, and scikit-learn, training classifiers on **10K+** data points with feature engineering
- Improved baseline accuracy by **15-25%** through hyperparameter tuning and model comparison against scikit-learn baselines

Software Engineer

First Tech Challenge Robotics | github.com/formula-r-ftc |

Jan 2020 - Jan 2022

Portland, OR

- Developed machine learning capabilities using object detection tools such as OpenCV, Vuforia, and TensorFlow to enhance robotic vision
- Programmed both TeleOp and autonomous robot functions, integrating rotation and light sensors to enable responsive and self-directed navigation

LEADERSHIP

Director of Website Development

Theta Tau

May 2025 - Jan 2026

Boston, MA

- Built and deployed a fully responsive website using HTML, CSS, and JavaScript tailored to the needs of the BU Engineering Fraternity
- Integrated a GitHub repository with Cloudflare Pages for seamless version control, automatic deployment, and real-time updates
- Configured custom domain, DNS settings, and SSL certificates through Cloudflare to ensure secure, stable, and high-performance web hosting

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, JavaScript, MATLAB, HTML/CSS, HDL, Bash

AI/ML & Computer Vision: PyTorch, TensorFlow, Hugging Face, Scikit-learn, OpenCV, NumPy, Pandas, CUDA, NCCL, Peretto, Roboflow, CVAT

Robotics & DevOps: ROS 1, ROS 2, Linux, OpenShift, Docker, Kubernetes, Git, GitHub Actions, Cloudflare, Jupyter, Vuforia, Motion Capture

CAD & Engineering Tools: NX, PTC Creo, SolidWorks, Onshape, PolyWorks